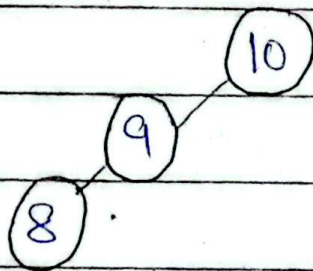


Topic: Rotations in AVL trees

Sub-topic: LL Rotation

Consider the sequence,
10, 9, 8

BST of given sequence is given by,

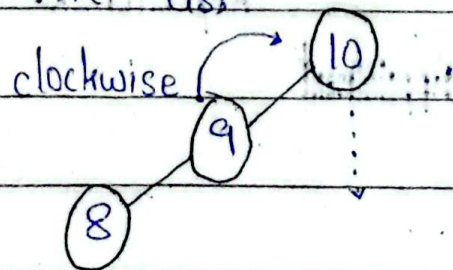


Balance factor of 8 node = 0

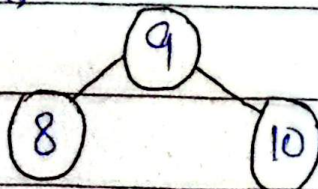
Balance factor of 9 node = 1

Balance factor of 10 node = 2

Rotate it as,



After rotation,



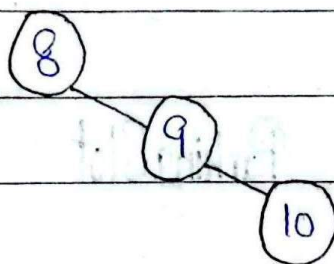
Topic: Rotations in AVL tree

Sub-topic: RR Rotation

Consider the sequence,

8, 9, 10.

BST of this sequence is given by,

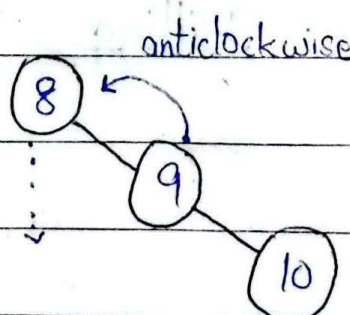


Balance factor of 10 node = 0

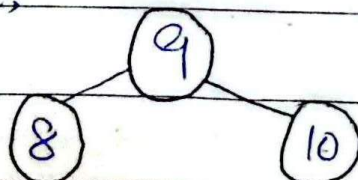
Balance factor of 9 node = -1

Balance factor of 8 node = -2

Rotate it as,



After rotation,



Date: _____

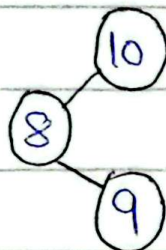
Day: _____

Topic: Rotations in AVL trees.

Sub-topic: LR Rotation

Consider the sequence,
10, 8, 9.

BST of given sequence is given by,



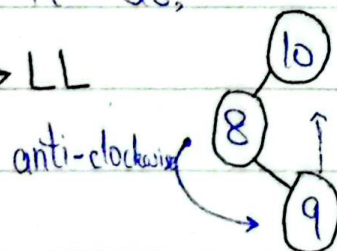
Balance factor of 9 node = 0

Balance factor of 8 node = -1

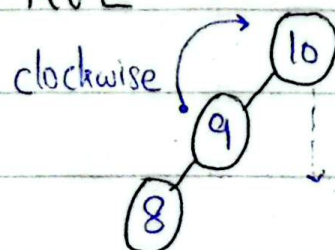
Balance factor of 10 node = 2

Rotate it as,

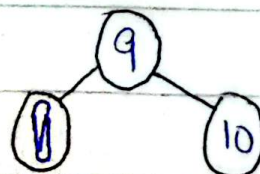
(i) LR $\rightarrow$ LL



(ii) LL $\rightarrow$ AVL



After rotation,



Date: _____

Day: _____

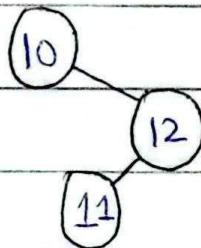
Topic: Rotations in AVL trees

Sub-topic: RL Rotation

Consider the sequence.

10, 12, 11.

BST of this given sequence is,



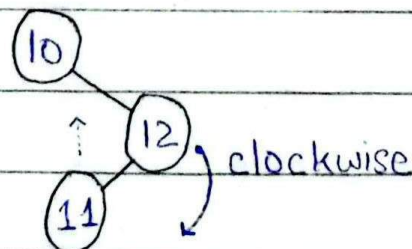
Balance factor of 11 node = 0

Balance factor of 12 node = 1

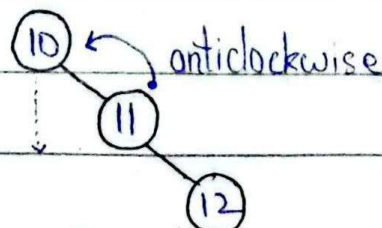
Balance factor of 10 node = -2

Rotate it as,

(i) RL $\rightarrow$ RR



(ii) RR $\rightarrow$ AVL



After rotations,

